Spatial Inequalities in Street-Tree Ecological Quality across New York City

Midterm Report

Urban Analytics

Track B (Advanced)

More Than Green
Di Li, Jialin Sun, Chunman Li, Xiaojing Jin

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1. Problem and Motivation

Urban street trees deliver a well-documented portfolio of ecosystem services: they mitigate the urban heat island effect, attenuate stormwater runoff, improve air quality, and contribute to residents' psychological well-being (Salmond et al., 2016; Berland et al., 2017). Despite growing consensus on their aggregate value, the spatial distribution of street-tree ecological quality is far from uniform. Evidence from North American cities consistently shows that lower-income, higher-density, and historically marginalized neighborhoods bear a disproportionate burden of degraded urban tree canopy (Wolf et al., 2020; Schwarz et al., 2015). New York City, with its extreme socioeconomic heterogeneity and dense built fabric, offers a compelling laboratory to examine these inequalities at fine spatial resolution.

This study pursues three interrelated research questions:

- RQ1: What spatial inequalities and clustering patterns characterize street-tree health, structural attributes, and species diversity across NYC?
- RQ2: How do street-tree ecological quality indicators associate with socioeconomic vulnerability, built-environment intensity (including 2D urban morphology and landscape configuration), and green infrastructure supply?
- RQ3: Where do multiple disadvantages overlap (low tree quality + high social vulnerability + high built density or fragmented greenery), and can we identify priority intervention areas?

Several key analytical constructs require explicit definition. The Tree Health Index is the mean of per-tree health scores (Good=3, Fair=2, Poor=1) aggregated to the Census Tract, providing a continuous proxy for overall structural condition. Shannon diversity (H') quantifies species richness weighted by evenness across the tract-level tree population. PLAND (proportion of landscape) expresses the percentage of a given land-cover class within each tract boundary, derived from the UrbanWatch 1m product. Landscape configuration is captured through a suite of FRAGSTATS metrics: Number of Patches (NP), Patch Density (PD), Largest Patch Index (LPI), Edge Density (ED), Aggregation Index (AI), Shannon Diversity Index (SHDI), and Contagion (CONTAG). Finally, co-burden mapping overlays multiple disadvantage layers to identify tracts simultaneously experiencing low tree quality, high social vulnerability, and high built density or fragmented greenery (Galle et al., 2021).

The primary stakeholders of this research include the NYC Department of Parks and Recreation (which administers the street-tree program), Community Boards seeking evidence for capital allocation, urban forestry

practitioners managing planting and maintenance programs, and environmental justice advocates who require spatially explicit co-burden evidence to support policy interventions.

The Census Tract is adopted as the unit of analysis (n = 2,327 tracts covering all five boroughs). This choice is justified on three grounds: (1) it permits direct integration with American Community Survey socioeconomic data; (2) it constitutes the standard spatial unit for environmental justice analyses in U.S. urban research; and (3) it offers sufficient spatial resolution to capture meaningful intra-city variation without the statistical instability associated with smaller block-group geometries.

Variable Domains and Indicators

Domain	Indicators
Tree Ecological Quality	Health Index, mean DBH, tree density (trees/km2), Shannon diversity (H'), species count
Built Environment (3D)	Mean/max building height, building coverage ratio, building density
Built Environment (2D morphology)	9-class PLAND from UrbanWatch 1m (building, road, parking, tree canopy, grass/shrub, water, agriculture, barren, other); impervious/pervious ratio
Landscape Configuration	SHDI, CONTAG (landscape-level); NP, PD, LPI, ED, AI for tree canopy and building classes
Green Infrastructure	Park area ratio, tree canopy PLAND
Socioeconomic	Median income, poverty rate, minority %, renter %, SVI composite (RPL_THEMES)

Table 1. Summary of analytical domains and indicators used in this study.

This framework draws on previous empirical work linking tree canopy cover to built-environment indicators (Berland et al., 2017), socioeconomic gradients (Wolf et al., 2020), and landscape configuration (Schwarz et al., 2015), while extending those foundations through the UrbanWatch high-resolution land-cover product (Zhang et al., 2022) and co-burden spatial methods (Galle et al., 2021).

2. Data Sources and Cleaning

Data Sources and Cleaning

Seven distinct datasets were assembled and harmonized to construct the tract-level analytical table. The following subsections describe each source's coverage, granularity, format, and provenance, followed by a record of cleaning and transformation decisions.

2.1 Trees Count! 2015 Street Tree Census

Source: NYC Open Data | Format: CSV with lat/lon | Coverage: City-wide, 2015-2016 |

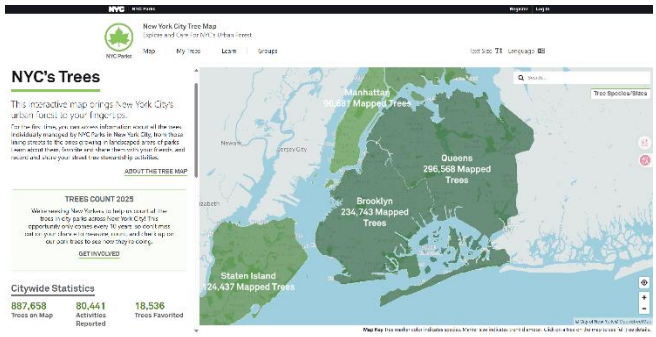
Raw records: 683,788

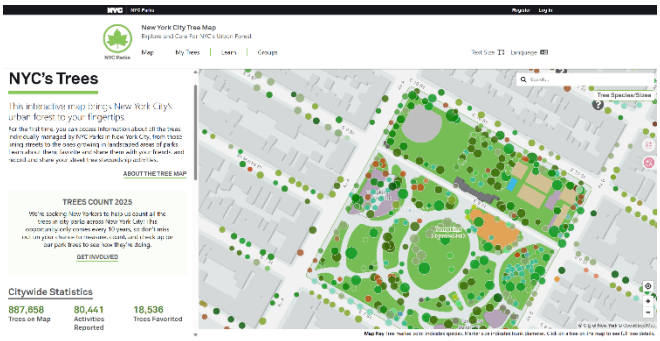
The NYC Parks Department conducted its third decennial street tree census between 2015 and 2016, recording detailed information for each mapped street tree across the five boroughs. The dataset includes species, diameter at breast height (DBH), health condition, and geographic coordinates, totaling 683,788 records in CSV format.

Data cleaning procedures first removed all records with a status other than “Alive,” resulting in 652,173 valid tree observations. Coordinate fields were checked to ensure completeness, with no missing or zero values identified. Implausible DBH values exceeding 100 inches were capped at 100 to mitigate the impact of

measurement errors. Tree health conditions were numerically encoded (Good = 3, Fair = 2, Poor = 1) to enable quantitative analysis.

Tree locations were then spatially joined to 2020 Census Tract boundaries (EPSG:4326), successfully matching 652,045 trees across 2,306 tracts. Aggregated tract-level indicators were computed, including total tree count, tree density (trees per square kilometer), mean DBH, a composite Tree Health Index, Shannon diversity index (H'), and species richness.

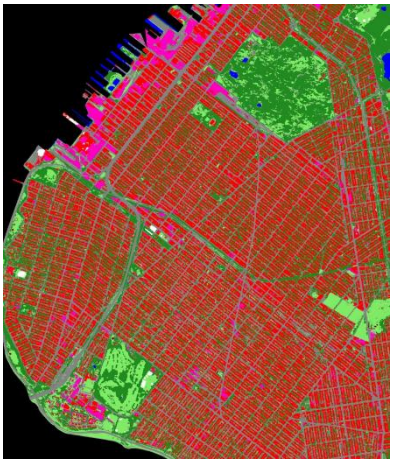




2.2 UrbanWatch 1m Land Cover

Source: University of Vermont / UNC Charlotte | Format: GeoTIFF (42,000 x 46,674 px) |

CRS: EPSG:26918 | Classes: 9



The UrbanWatch dataset provides a high-resolution (1-meter) land cover classification for the greater NYC area, derived from 2017 aerial imagery and validated at over 90% overall accuracy. The dataset is distributed as a three-band RGB GeoTIFF, where each RGB combination corresponds to one of nine land cover classes.

To enable analysis, the RGB raster was decoded into a single-band classified raster using the provided class-color lookup table, resulting in approximately 749 million valid pixels. Census Tract boundaries were reprojected to UTM Zone 18N (EPSG:26918) to align with the raster data. Zonal statistics were then computed to extract pixel counts per land cover class within each tract.

From these results, PLAND metrics were derived, representing the percentage of each land cover class relative to total tract area, yielding ten indicators per tract. In addition, landscape configuration metrics—including Number of Patches (NP), Patch Density (PD), Largest Patch Index (LPI), Edge Density (ED), Aggregation Index (AI), Shannon Diversity Index (SHDI), and Contagion Index (CONTAG)—were calculated using a FRAGSTATS-equivalent Python workflow at the tract level.

P.S. FRAGSTATS: FRAGSTATS is a landscape analysis tool used to quantify spatial patterns in categorical

raster data. It identifies contiguous pixels of the same class as patches and computes metrics describing fragmentation, connectivity, edge complexity, and spatial heterogeneity. These metrics are widely applied in landscape ecology and urban studies to understand spatial structure.

In this study, FRAGSTATS-based metrics complement Python-based geospatial analysis. While Python is effective for calculating land-cover composition (e.g., PLAND), it does not provide a standardized framework for analyzing spatial configuration. FRAGSTATS offers a well-established set of patch-based metrics that ensure methodological consistency and comparability.

2.3 NYC Building Footprints

Source: NYC DoITT / NYC Open Data | Format: GeoPackage | Records: 1,082,289 |

CRS: EPSG:4326

The NYC Building Footprints dataset, provided by NYC DoITT, contains polygon geometries for all mapped buildings across the city, along with an attribute for roof height in feet. The dataset includes 1,082,289 records in GeoPackage format.

For processing, building geometries were reprojected to UTM Zone 18N to ensure accurate area calculations. The height_roof attribute was converted from feet to meters using a conversion factor of 0.3048. Representative centroid points were generated for each building and spatially joined to Census Tracts, resulting in a successful match rate of 99.98% (1,082,092 buildings).

Tract-level indicators derived from this dataset include total building count, mean, median, and maximum building height (in meters), building footprint coverage ratio (total footprint area divided by tract area), and building density (buildings per square kilometer). A small number of tracts corresponding to water bodies or major park interiors contained no buildings and were assigned missing values.

2.4 PLUTO v25.4

Source: NYC Department of City Planning | Format: CSV | Records: 858,644 tax lots

MapPLUTO v25.4, published by the NYC Department of City Planning, provides parcel-level land use and building attributes for 858,644 tax lots in CSV format. Key variables include floor area ratio (FAR), number of floors, year built, and land use classification.

To integrate this dataset, a consistent 11-digit Census Tract GEOID was constructed by concatenating the borough code (borocode) and tract identifier (bct2020). Parcel-level attributes were then aggregated to the tract level. Derived indicators include mean FAR, mean number of floors, median year built, and the modal land use class within each tract.

2.5 NYC Parks Properties

Source: NYC Parks / NYC Open Data | Format: CSV with WKT geometry | Records: 2,058

The NYC Parks Properties dataset enumerates all NYC Parks Department-managed properties, represented as multipolygon geometries stored in WKT format. The dataset includes 2,058 records.

WKT geometries were parsed using the Shapely library and intersected with Census Tract boundaries to compute the proportion of each tract occupied by parkland. This resulted in a park area ratio indicator for each tract. Of the 2,327 tracts, 1,354 contained at least partial park coverage, while the remaining 973 tracts were assigned a park_area_ratio of zero. These zero values were treated as structurally missing in terms of park presence rather than absent data.

2.6 American Community Survey (ACS) 5-Year Estimates (2019)

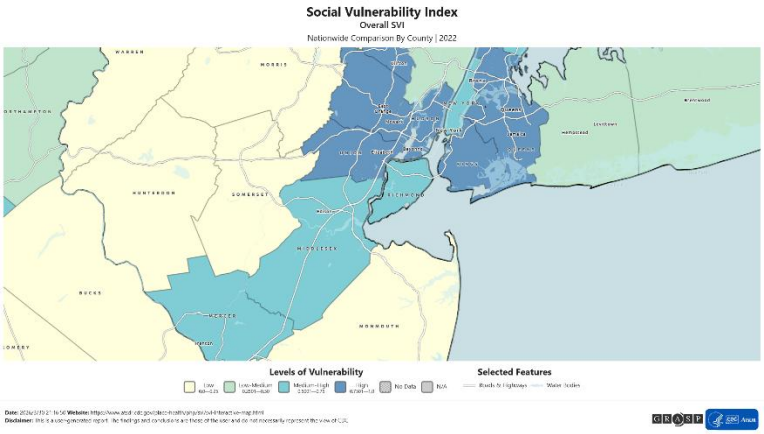
Source: U.S. Census Bureau API | Coverage: 2,167 NYC tracts | Tables: B19013, B17001, B02001, B25003

Socioeconomic variables were obtained from the U.S. Census Bureau's American Community Survey (ACS) 2019 5-year estimates via the Census API, covering 2,167 NYC Census Tracts. Four tables were used: B19013 (median household income), B17001 (poverty status), B02001 (race), and B25003 (housing tenure).

From these tables, key indicators were computed, including poverty rate (percentage of population below the poverty line), minority percentage (non-white population share), and renter percentage (share of renter-occupied housing units). Sentinel values (-666666666) used by the Census Bureau to denote missing or suppressed data were recoded as NaN. Approximately 160 tracts (7%) had suppressed median income estimates, primarily corresponding to small or institutionalized populations.

2.7 CDC/ATSDR Social Vulnerability Index (SVI)

Source: CDC / ATSDR | Format: CSV | Coverage: All U.S. Census Tracts



The CDC/ATSDR Social Vulnerability Index (SVI) dataset provides a composite measure of social vulnerability at the Census Tract level across the United States. The primary indicator, RPL_THEMES, ranges from 0 to 1 and summarizes vulnerability across four dimensions: socioeconomic status, household composition, minority status and language, and housing type and transportation.

SVI data were matched to the analytical dataset using the 11-digit Census Tract GEOID. Sentinel values (-999) were recoded as NaN, affecting approximately 347 tracts (14.9%), which largely correspond to unpopulated areas such as water bodies, parks, or institutional zones identified by the CDC.

2.8 Final Data Integration

All seven datasets were integrated at the Census Tract level using the GEOID as a common key. The final analytical dataset consists of 2,327 Census Tracts and 73 variables, capturing a comprehensive range of environmental, spatial, and socioeconomic characteristics.

To ensure methodological consistency, all spatial operations—including reprojection, area calculation, and spatial joins—were conducted using a unified projected coordinate system, EPSG:26918 (UTM Zone 18N, NAD83). This standardization is critical for maintaining accuracy in distance- and area-based metrics across datasets with differing original coordinate systems.

3. Missing Data Analysis

3.1 Missing Data Identification

To evaluate data integrity, we systematically examined the distribution of missing values across the final tract-level dataset. Missingness was analyzed not only in terms of magnitude but also in terms of its structural and spatial context.
We distinguished between structural absence and data gaps.
This distinction is critical for applying the Dark Data framework, as it separates meaningful “known missing” values that reflect real-world conditions from missingness arising from data collection or reporting limitations.

3.2 Missing Data Identification

Missing data in the analytical table are structurally informative rather than random. Following the dark data taxonomy (Hand, 2020), each variable group is classified by missingness mechanism and handled accordingly.

Variable Group	Missing N (%)	Dark Data Type	Handling Strategy
Tree indicators (health, diversity)	21 tracts (0.9%)	DD-Type 2 (Known missing)	tree_count = 0 assigned; diversity = NaN for tracts with < 5 trees
Building indicators	5 tracts (0.2%)	DD-Type 2 (Known missing)	Retained as NaN (water / major park tracts)
Park area ratio	973 tracts (41.8%)	DD-Type 2 (Known missing)	Filled with 0.0 (no parkland present)
Median income	~160 tracts (~6.9%)	DD-Type 3 (Choosing not to record)	Retained as NaN; listwise deletion applied in regression models
SVI RPL_THEMES	~347 tracts (~14.9%)	DD-Type 5 (Measurement issues)	Recoded -999 to NaN; excluded from SVI-dependent analyses
Landscape metrics	7 tracts (0.3%)	DD-Type 4 (Self-selection / edge)	Retained as NaN (tracts at raster boundary)

Table 2. Missing data summary by variable group, mechanism classification, and handling strategy. The missing data matrix is shown in Figure 6.

3.3 Spatial and Structural Patterns of Missingness

Missing data are not randomly distributed but instead exhibit clear structural and spatial patterns.

Tree-related missingness is concentrated in tracts with very low tree counts, reflecting insufficient observations rather than data collection failure. Additionally, some apparent absence of tree-related indicators results from differences in data definitions rather than true ecological absence. The TreesCount dataset records only street trees and excludes trees located within parks or other non-street environments. Consequently, areas such as Central Park may appear to have zero tree counts despite substantial tree canopy, indicating a limitation of data scope rather than actual absence.

Park area ratio shows a high proportion of missing values (41.8%); however, this reflects a structural condition, as most tracts do not contain NYC Parks-managed land. These cases are therefore recoded as zero to accurately represent the absence of parkland.

Socioeconomic variables, particularly median income, exhibit missingness due to data suppression in low-population tracts, which are often associated with institutional land uses or minimal residential presence. Similarly, missingness in the Social Vulnerability Index (SVI) is primarily driven by a sentinel value (-999), which systematically excludes non-residential or institutional tracts.

3.4 Implications for Bias and Analysis

The structured nature of missing data has important implications for analysis.

Because SVI and income data are missing primarily in non-residential or low-population tracts, excluding these observations from regression models is analytically appropriate and unlikely to introduce systematic bias into comparisons across residential neighborhoods. However, such exclusion may slightly underrepresent areas with atypical land use characteristics.

The recoding of park area ratio to zero ensures that the absence of parkland is correctly represented rather than misinterpreted as missing data. For tree-related indicators, assigning tree_count = 0 while retaining undefined

values (*NaN*) for diversity preserves the distinction between tracts with no trees and those with measurable ecological variation.

To further ensure robustness, a sensitivity analysis will be conducted by excluding tracts with suppressed income data from all income-based regressions.

Overall, although missingness is substantial in certain variables, its structured nature allows for analytically appropriate handling with limited risk of introducing bias into the main findings.

4. Initial Results and Visualizations

4.1 Descriptive Summary

After cleaning and merging, the analytical dataset captures 652,045 alive street trees distributed across 2,306 of 2,327 Census Tracts. Summary statistics for key indicators are presented below.

Indicator	Mean	Std. Dev.	Min	Max
Tree Health Index	2.76	0.12	1.25	3.00
Tree Density (trees/km2)	1,103	712	0	4,890
Mean DBH (inches)	11.4	3.2	1.0	100.0
Shannon Diversity (H')	1.82	0.44	0.00	2.89
Species Count per Tract	28.3	14.1	1	82
Tree Canopy PLAND (%)	27.4	13.5	0.3	71.2
Impervious PLAND (%)	67.0	14.8	11.2	97.8
Mean Building Height (m)	11.6	9.3	0.0	130.4
Park Area Ratio (%)	7.3	14.2	0.0	98.4
Median Household Income ($)	69,420	38,110	9,800	250,001
Poverty Rate (%)	17.3	14.2	0.0	71.8
Minority %	56.4	30.1	1.2	99.8
SVI RPL_THEMES	0.51	0.28	0.00	1.00

Table 3. Descriptive statistics for key tract-level indicators (n = 2,306 tracts with tree data).

Health condition composition across all alive trees is: Good 81.1%, Fair 14.8%, Poor 4.1%. The top three street-tree species by count are London planetree (18.3%), honeylocust (9.9%), and Callery pear (9.0%), the latter raising long-term ecological concerns given its invasive potential. Native species constitute a small minority of the total inventory, concentrated in lower-ranked species by abundance.

4.2 Spatial Distribution of Tree Health Index

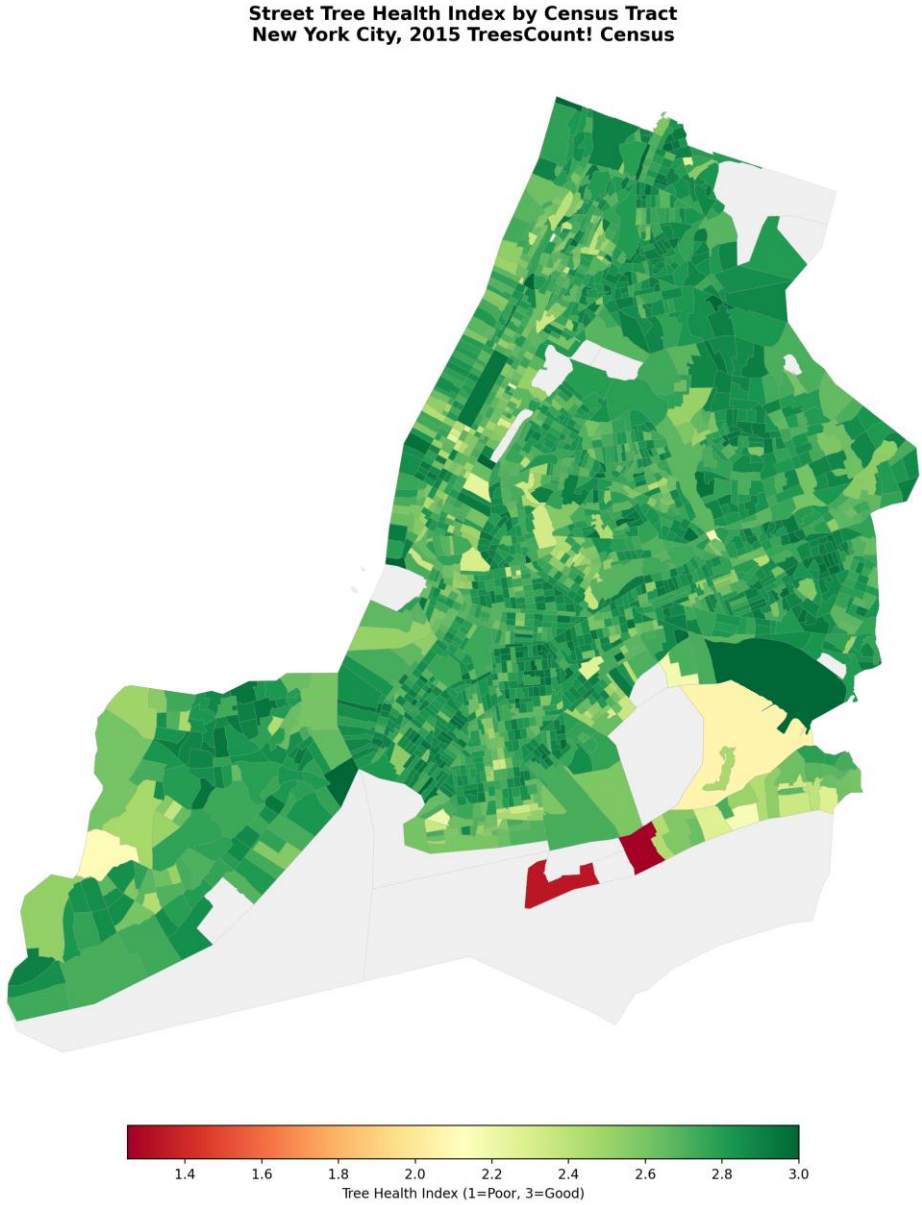


Figure 1. Choropleth Map of Street Tree Health Index by Census Tract New York City, 2015 TreeCount! Census

Figure 1 presents a choropleth map of the Tree Health Index across all 2,306 tracts. The citywide pattern reveals a predominantly green landscape with mean health near 2.76, consistent with a majority-Good distribution. However, pronounced low-health clusters are visible in lower Manhattan (Financial District, Lower East Side) and along the south Brooklyn and south Queens coastlines. These areas combine high building density, elevated impervious surface, and lower canopy coverage, suggesting built-environment constraints on tree vigor. Staten Island exhibits the highest health scores city-wide (see Figure 2).

4.3 Borough-Level Health Composition

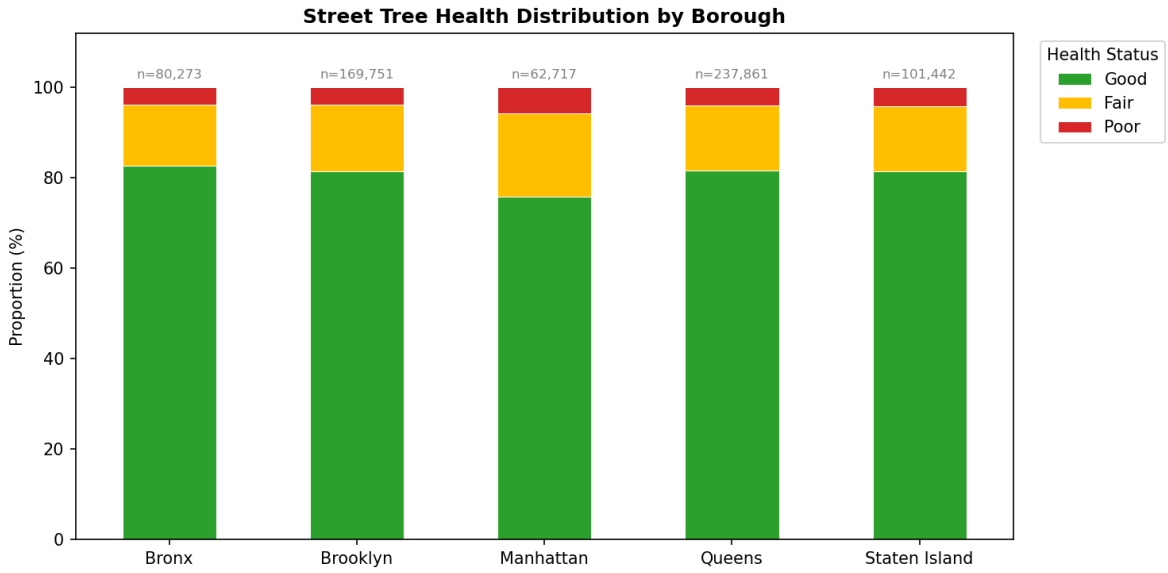


Figure 2. Stacked Bar Chart of Street Tree Health Distribution by Borough

Figure 2 disaggregates health composition by borough. Manhattan shows the lowest proportion of Good-rated trees (~76%) and the highest Poor proportion (~5%), despite having the highest median income among its resident populations. This apparent contradiction is consistent with the built-environment hypothesis: Manhattan's dense, impervious streetscape constrains root zones and moisture retention regardless of neighborhood wealth. Staten Island has the highest Good proportion and lowest Poor proportion, reflecting lower building density and greater canopy coverage.

4.4 Species Composition

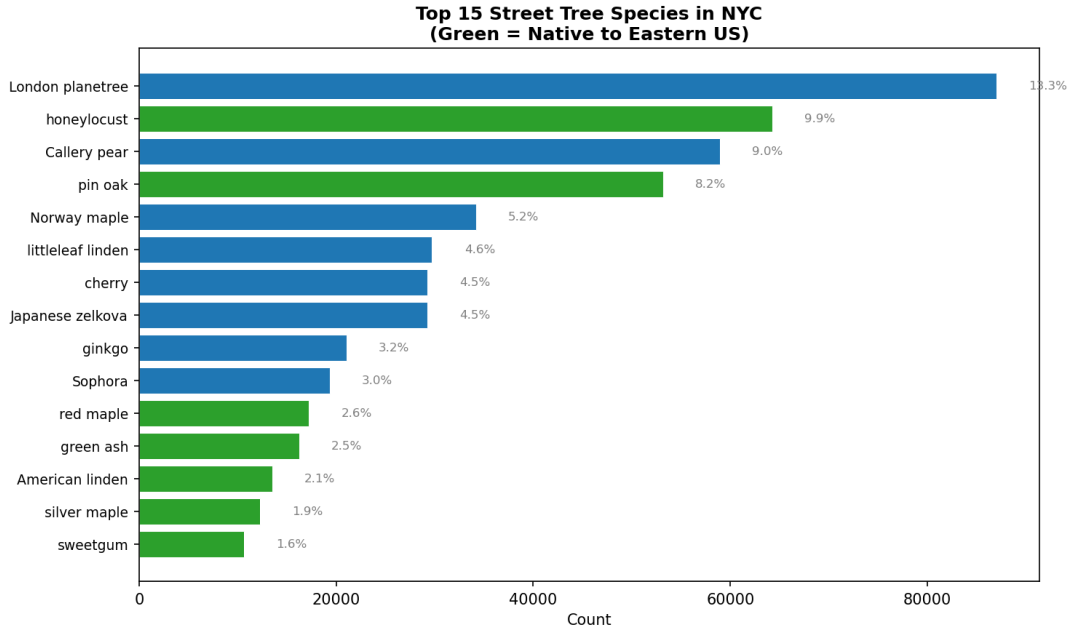


Figure 3. Horizontal Bar Chart of Top 15 Street Tree Species in NYC

Figure 3 displays the top 15 street-tree species by count. London planetree dominates, comprising nearly one-fifth of all alive trees — a legacy of mid-20th century mass planting driven by its pollution tolerance. The predominance of three non-native species in the top ranks (London planetree, honeylocust, Callery pear) underscores ongoing concerns about species homogeneity and its implications for ecological resilience under pest or climate pressures (Berland et al., 2017).

4.5 Socioeconomic Correlates

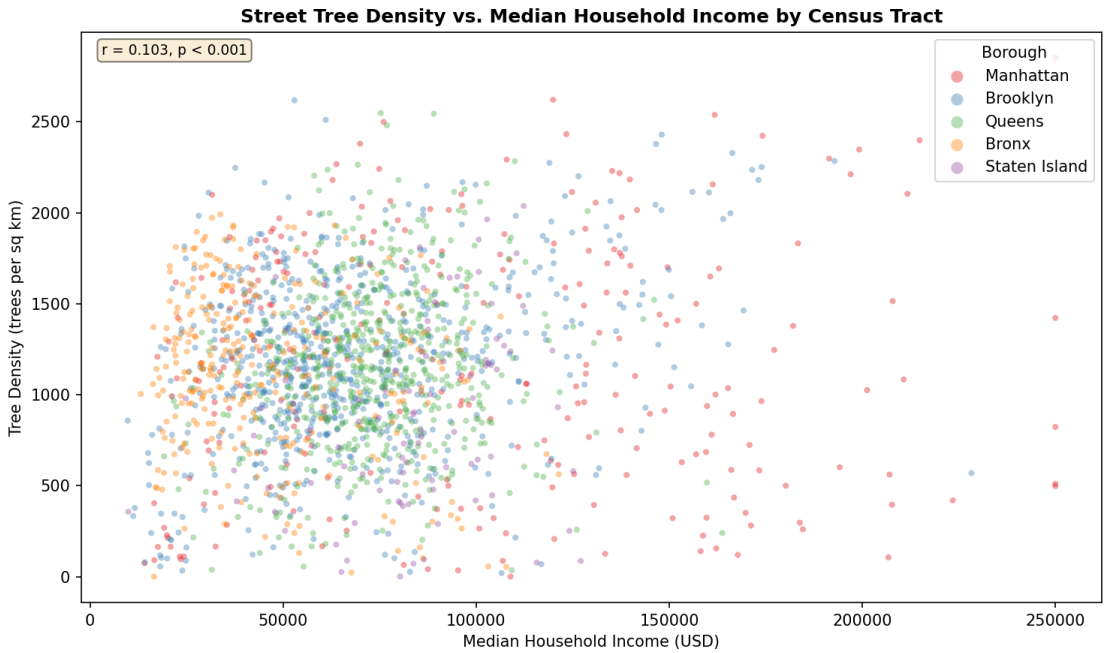


Figure 4. Scatter Plot of Street Tree Density vs. Median Household Income

Figure 4 examines the bivariate relationship between tree density and median household income. A statistically significant but weak positive correlation is observed ($r = 0.102$, $p < 0.001$), indicating that higher-income tracts tend to have marginally higher tree density. Manhattan tracts form a visually distinct cluster characterized by high income but anomalously low tree density, reflecting the borough's extreme built-environment intensity. This pattern suggests that income alone is insufficient to explain tree distribution; built-environment constraints must be considered jointly.

4.6 Land Cover Composition by Borough

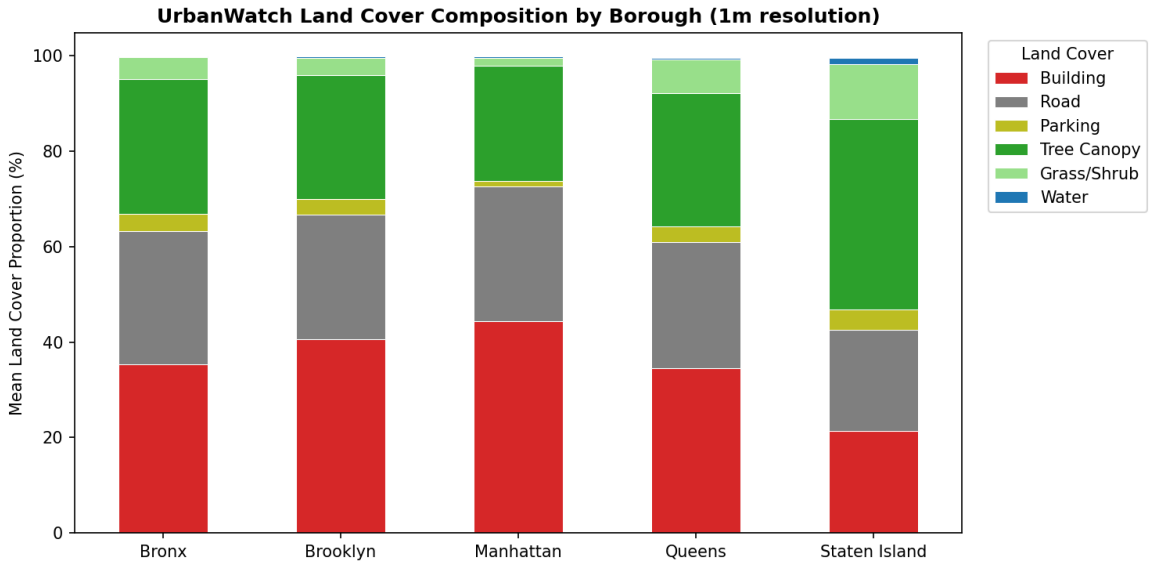


Figure 5. Grouped bar chart of Land Cover Composition by Borough(1m resolution)

Figure 5 presents land cover composition (PLAND by class) stratified by borough. Staten Island has the highest mean tree canopy PLAND (~40%) and lowest impervious surface (~52%), while Manhattan has the lowest tree canopy (~6%) and highest impervious cover (~93%). The Bronx and Brooklyn occupy intermediate positions, with Brooklyn showing notable variability across tracts. These borough-level patterns establish the structural context within which tree health and diversity outcomes are subsequently modeled.

4.7 Built Environment Intensity and Tree Health

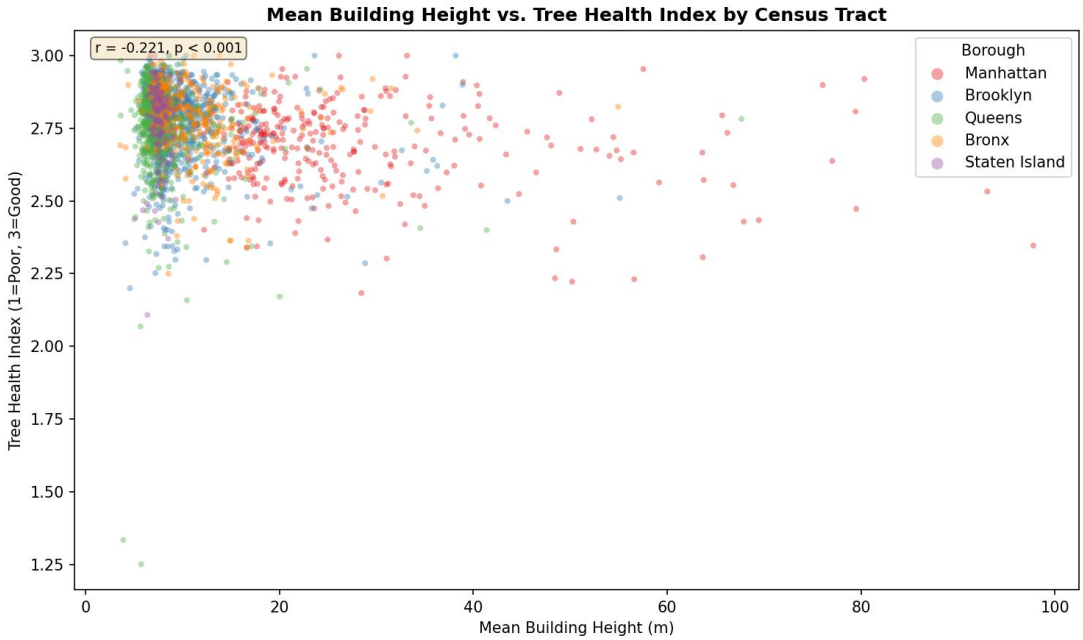


Figure 6. Scatter Plot of Mean Building Height vs. Tree Health Index

Figure 6 examines the relationship between mean building height and Tree Health Index at the tract level. A significant negative correlation is observed ($r = -0.221$, $p < 0.001$), indicating that tracts with taller buildings tend

to exhibit lower tree health. This effect is consistent with the hypothesis that vertical densification reduces solar access, increases street-canyon wind effects, and compresses root-zone soil volumes available to street trees. The relationship is notably stronger in the Pearson correlation than the income-tree density relationship, suggesting that physical built-environment factors may be more proximal determinants of tree condition than socioeconomic ones.

4.8 Caveats and Limitations of Initial Results

- The tree census is point-in-time (2015-2016); tree conditions observed may not reflect current conditions due to subsequent plantings, removals, storm damage, and aging.
- Health condition is assessed categorically by trained volunteers; inter-rater variability and the coarse three-level scale limit sensitivity to gradations of tree stress.
- UrbanWatch (2017 imagery) and the tree census (2015-2016) have a partial temporal offset; modest land-cover change between these windows could introduce measurement error in PLAND-health correlations.
- All bivariate correlations reported here are descriptive; confounding is extensive in the urban landscape and correlation does not imply causation.

5. Progress Since Proposal and Remaining Plan

5.1 Changes from the Original Proposal

- iNaturalist removed: The iNaturalist citizen-science occurrence data were originally proposed as a supplementary biodiversity layer. Upon inspection, the dataset exhibited substantial spatial sampling bias toward parks and high-foot-traffic areas, making tract-level aggregation misleading. The dataset was dropped in favor of the TreesCount! census species records, which provide complete and unbiased coverage of the street-tree population.
- Building height source changed: LiDAR-derived building heights were proposed in the original submission. The NYC Building Footprints dataset, which includes the height_roof attribute derived from photogrammetric surveys, provides equivalent vertical structure information with substantially lower data access friction and better coverage (1,082,289 buildings).
- UrbanWatch landscape configuration added: The landscape metrics dimension (FRAGSTATS-equivalent parameters from UrbanWatch 1m classification) was not in the original proposal but has been incorporated as a major analytical dimension, providing 14 configuration parameters per tract that quantify patch complexity, connectivity, and landscape heterogeneity independently from simple PLAND coverage.
- RQ scope refined: Three original RQs were reorganized to the present structure (RQ1 spatial patterns, RQ2 associations, RQ3 co-burden) following the removal of the iNaturalist layer, which had supported an originally planned fourth RQ on wildlife co-occurrence.

5.2 Work Completed

- All seven data sources acquired, documented, and version-controlled.
- Full data cleaning pipelines implemented and logged (see Section 2).
- Tract-level indicator table constructed: 2,327 tracts x 73 variables.
- UrbanWatch RGB decoded to single-band classified raster; zonal stats computed for all tracts.
- Landscape configuration metrics (14 FRAGSTATS parameters) computed via Python FRAGSTATS equivalent workflow.
- Descriptive statistics, health composition analysis, and six core visualizations completed.

- Missing data audit completed and dark-data classification assigned (Section 3).

5.3 Remaining Work and Timeline

Week	Tasks
Week 4	Spatial autocorrelation: global Moran's I for Health Index, diversity, and density; local LISA cluster maps (HH/LL/HL/LH) for each tree quality indicator.
Week 5	Regression modeling and variable selection: Preliminary screening of 73 variables to address multicollinearity and improve model interpretability (Correlation matrix analysis, Variance Inflation Factor (VIF) filtering). Stepwise (hierarchical) model construction: **Model 1:** socioeconomic variables only, **Model 2:** + built environment variables, **Model 3:** + landscape configuration variables. Comparison of model performance to evaluate incremental explanatory power and identify dominant factor groups. GLMs with Health Index and Shannon diversity as outcomes. Spatial lag or spatial error correction if residual autocorrelation is significant. Co-burden overlay: composite disadvantage index combining low tree quality + high SVI + high built density or fragmented greenery
Week 6	Interactive HTML maps via Folium or Mapbox GL JS for LISA clusters and co-burden spatial distribution. GitHub Pages deployment. Final written report assembly and figure panel finalization.

Table 4. Remaining task timeline through project completion.

6. References

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